# Management Theory and Practice

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# Topics to cover this week

1. Overview
2. Management theory
3. Management functions
4. Organization
5. Project overview



# Self-introduction

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# Week 2-6

- **Week 2** - Overview, management theory and practice
- **Week 3** - Procurement strategy
- **Week 4** - Tendering approach and measurement
- **Week 5** - Contract, risk, and quality management
- **Week 6** - Group project presentation (bidding)



# Project overview

**Project goal:** In groups of $\sim$ 5 people, identify a sustainability-related campus improvement opportunity and develop a tender for the proposed project.

**Roles and tasks**: Project coordinator, Scope & Sustainability, Procurement & Tendering, Contract & Commercial, Quality & Delivery
**Deliverables**: A concise tender package and a group presentation in Week 6

## Assessment

The project is assessed on the quality and coherence of your project-management and procurement decisions, **NOT** on advanced engineering design.

Could HKU Estates Office use your tender to obtain and evaluate credible bids for this retrofit project?

# Project timeline

- **Week 2** - Overview, management theory and practice
  Team formation, role assignment, background research
- **Week 3** - Procurement strategy
  Scope definition, route/method selection (in-class discussion)
- **Week 4** - Tendering approach and measurement
  Tender requirements, pricing, and evaluation criteria
- **Week 5** - Contract, risk, and quality management
  Contracts, risk register, and quality plan (in-class discussion)
- **Week 6** - Group project presentation (bidding)
  GROUP presentation, package submission (Week 7)



# Think about a daily life example

**What is (project) management?**



# What is management?

The process of **coordinating** people, resources, and activities to achieve defined organizational **objectives**.

- Everywhere: government, university, department, contractor, etc.
- Objectives → Planning → Organizing → Execution → Monitoring & Control → Adjustment
- *What is NOT management?*



# What defines a project?

Temporary undertaking created to deliver a defined outcome in a unique context.

- Specific objective or deliverable
- Defined time frame
- Limited resources
- Interdependent activities
- Multiple participants and stakeholders
- Uncertainty and risk

| Operations | Project |
|---|---|
| Repetitive activities | Unique outcome |
| Ongoing | Temporary |
| Continuous | Defined completion |
| Stable organization | Changing resources |
| Standard processes | Higher uncertainty |



# Project management

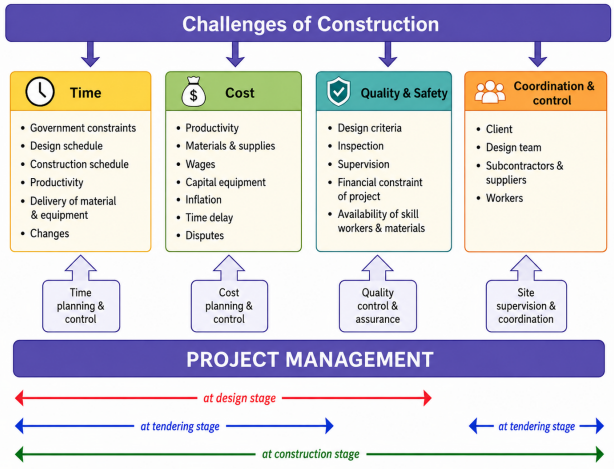
Application of management principles, processes, and tools to deliver a project.

- **Scope** - What must be delivered?
- **Time** - When must it be completed?
- **Cost** - What resources can be spent?
- **Quality** - What performance is required?
- People and organizations
- Risk and uncertainty
- Information and communication

**Interdependent** decision-making and resource coordination under constraints.



# PM in civil engineering




- Technical complexity
- Organizational complexity
- Uncertainty
- External constraints
- Long time horizon

*A construction project finishes six months late and 20% over budget. Who has failed, and why?*

# Recap

**Interdependent** decision-making and resource coordination under constraints.



# Interconnected instead of sequential activities



# Planning - what are we trying to achieve?

Define the desired outcome and determine how to achieve it.

- Define project objectives
- Scope and deliverables
- Break work into activities
- Identify required resources
- Establish sequence and schedule
- Estimate costs and budget
- Identify major risks and assumptions



# Planning - from objectives to actions

Example of campus HVAC retrofit:

- Define project objectives - Reduce cooling energy consumption by 25%
- Scope and deliverables - Replace selected AHUs and pumps
- Break work into activities - Procurement, installation, commissioning, etc.
- Identify required resources - Technician, sensors and controls, room access, etc.
- Establish sequence and schedule - Construction duration: 12 weeks
- Estimate costs and budget - Budget: HK$8 million
- Identify major risks and constraints - Maintain indoor conditions during operating hours



# Planning - decisions at different levels

It goes beyond designing a high-level program.

| Level | Typical question | Example |
|---|---|---|
| Strategic | What should we do? | Retrofit existing building or construct replacement? |
| *Project/tactical* | How should the project be delivered? | Phasing, procurement strategy, major milestones, etc. |
| Operational | What happens next? | Which area to work on next Monday? |

*Good planning reduces uncertainty and operational friction during project execution.*



# Organizing - who does what?

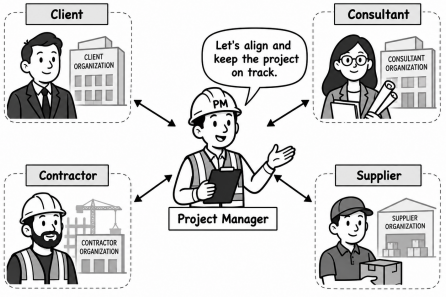
Translate the project plan into a structure to deliver it.

- Divide work into manageable packages
- Assign responsibilities: obligation to perform a task
- Allocate resources
- Establish authority: ability to make decisions
- Define reporting relationships
- Establish coordination mechanisms
- Define communication channels

Work → Responsibility → Authority → Coordination



# Organizing - multi-organizational coordination




PMs frequently coordinate people who do not report directly to them.

PMs coordinate across organizational interfaces, where the work, responsibility, or information of two parties meet.

# Leading - how to get people to work together?

Leading the people, not the tasks.

- Communicate objectives and priorities
- Make decisions and give directions
- Coordinate team members
- Motivate people
- Resolve disagreements
- Negotiate between competing interests
- Build collaboration

Leadership addresses different objectives from stakeholders/participants, through alignment and negotiation.



# Leading - Communication? Communication. Communication!

Communication for information management: what information was issued, when, to whom, and whether the change was acknowledged and understood

- E.g. revised drawing issued → formal change notice → affected parties identified → contractor confirms receipt → impact on procurement and program discussed → obsolete drawing withdrawn

*If everyone communicated clearly and on time, why is there still conflict?*



# Controlling - are we achieving the plan?

Set target → Measure actual performance → Compare → Identify deviation → Take corrective action

| Area | Planned | Actual | Management response |
|---|---|---|---|
| Schedule | 40% complete | 32% | Re-arrange work/add resources |
| Cost | HK$4M | HK$4.6M | Investigate variance |
| Quality | $\leq$2% defects | 6% | Identify root cause/corrective action |
| Safety | Zero incidents | Incident occurs | Investigation and control measures |

Monitoring relies on quantifiable planning.



# Controlling - monitor and then react

E.g. Project is two weeks behind schedule.

**Monitoring**: report the two-week delay

**Controlling**:

- Identify which activities caused the delay
- determine whether they affect the critical program
- assess available responses
- reallocate resources or re-sequence work
- update the program
- monitor whether the intervention works

*Close the functional cycle*



# Controlling - management by exception

*Should a project manager try to eliminate every deviation from the plan?*
Identify deviation → Diagnosis → Take corrective action

Delay → X → add more workers

- Other causes such as design issue, material delay, etc.
- Potential consequences such as additional cost/equipment, overtime fatigue, etc.

Adjustment should not only involve planning, but the entire management cycle.



# Recap

**Scenario**: During a campus retrofit, complaints show that construction noise is disrupting teaching.

- **Planning**: What should we change?
- **Organizing**: Who needs to react?
- **Leading**: How do we achieve agreement and cooperation?
- **Controlling**: What should be monitored after adjustment?



# Recap

**Scenario**: During a campus retrofit, complaints show that construction noise is disrupting teaching.

- **Planning**: Can noisy work be rescheduled? What effect will that have on time and cost? Other classrooms available?
- **Organizing**: Who checks teaching schedules? Who approves work windows? Who coordinates access and informs affected users?
- **Leading**: Were users and contractors properly informed? What compromise can satisfy the main constraints?
- **Controlling**: Has the conflict been resolved? Is the process documented? Is there a new conflict? Is the budget/timeline affected?



# Organization - how should work be divided?

Projects are too complex for one person or one team to deliver everything. Work therefore needs to be divided by:

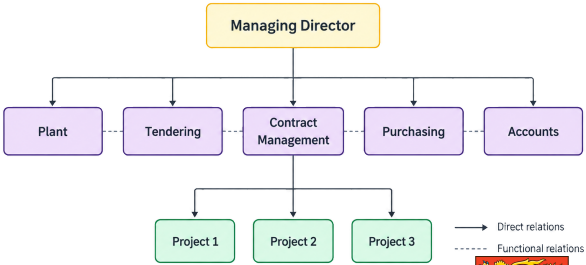
- **Discipline**: architectural, structural, geotechnical, MEP
- **Activity**: design, procurement, installation, testing
- **Location**: Building A, Building B, Zone 1, Zone 2
- **System/package**: facade, HVAC, electrical, lifts
- **Project phase**: design, construction, commissioning

Specialization improves efficiency, but creates dependencies between people and teams. *How should people be grouped?*



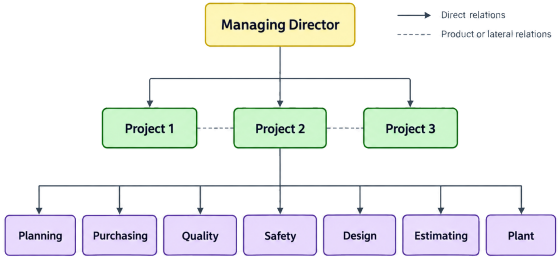
# Organization - functional structure

- ✓ Strong technical specialization
- ✓ Clear professional development and technical supervision
- ✓ Efficient sharing of specialist staff
- × Project priorities overtaken by departmental priorities
- × Slower decisions across disciplines
- × Weaker end-to-end project ownership



# Organization - project-based structure

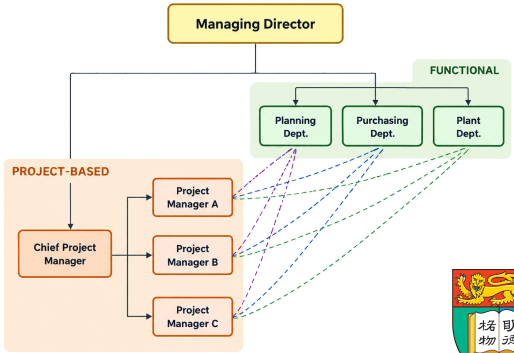
- ✓ Clear project focus
- ✓ Stronger PM authority
- ✓ Faster project-level decisions
- × Duplication of specialists across projects
- × Less expertise to preserve quality
- × Staff deployment after project completion



# Organization - matrix structure

Functional ← Engineer → Project
Manager Manager

- ✓ More efficient expertise utilization
- ✓ Strengthened project coordination compared with a purely functional structure
- × Competition for resources and difficulty in defining authority



# Organization - structural coordination mechanisms

| **Structural mechanism** | **What it does** |
|---|---|
| *Reporting relationships* | Establish who supervises whom and where formal accountability sits |
| *Decision rights and authority* | Define who can approve, instruct, commit resources, or change the work |
| *Liaison roles* | Assign people to connect teams or organizations |
| *Coordination meetings* | Create a forum for resolving interdependent issues |
| *Approval routes* | Specify how decisions requiring formal authorization are processed |
| *Escalation paths* | Define where unresolved issues go when existing authority is insufficient |



*Start working on your project!*

